

CITATIONFLOW ANALYTICS

Research Citation Network Analysis and Influence Ranking System

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ABSTRACT

- CitationFlow Analytics analyzes relationships between research papers.
 - Each research paper is represented as a node.
 - A citation between two papers is represented as a directed edge.
 - The system builds a citation network to study the flow of knowledge.
 - It identifies influential research papers based on citation relationships.
 - It also visualizes citation paths and generates analytical reports.
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PROBLEM STATEMENT

- There are a large number of research papers published every year.
- Understanding the relationships between papers manually is difficult.
- It is hard to identify influential research papers in a large citation network.
- Tracking how one research paper influences other papers can be complex.
- A system is needed to efficiently analyze citation relationships and knowledge flow.

PROPOSED FUNCTIONALITIES

The system will provide:

- Add and manage research papers
- Create citation relationships between papers
- Build a directed citation graph
- Identify influential research papers
- Analyze citation paths
- Calculate citation flow and paper rankings
- Visualize the citation network
- Generate citation analysis reports

Tools and Technologies

- Java
- Directed Graph
- Adjacency List
- BFS
- DFS
- Edmonds-Karp (Maximum Flow)
- Priority Queue / Sorting
- Java Collections Framework
- HTML
- CSS
- JavaScript
- Java HttpServer

Expected Outcome

- Create a network of research papers and citations
- Represent citation relationships using a directed graph.
- Identify important and influential research papers.
- Show how knowledge flows through the citation network.
- Display citation paths and relationships visually.
- Generate rankings and analytical reports.
- Efficiently handle multiple research papers and citation relationships.

PROJECT IMPLEMENTATION:



CitationFlow Analytics

Research Citation Network Analysis & Influence Ranking System

DSA • Directed Graph • Network Flow

8

Research papers (V)

15

Citation edges (E)

P1

Top paper by in-degree

3

Demo maximum flow

$O(VE^2)$

Edmonds-Karp

01 • MODEL

Directed Citation Graph

A research paper is a vertex. A citation is a directed edge from the **citing** paper to the **referenced** paper. Every edge has a positive capacity for the network-flow model.



ImageNet Classification with Deep Convolutional Neural Networks → ImageNet: A large-scale hierarchical image database
capacity = 1



Deep Residual Learning for Image Recognition → ImageNet Classification with Deep Convolutional Neural Networks
capacity = 1



Deep Residual Learning for Image Recognition → Very Deep Convolutional Networks for Large-Scale Image Recognition
capacity = 1

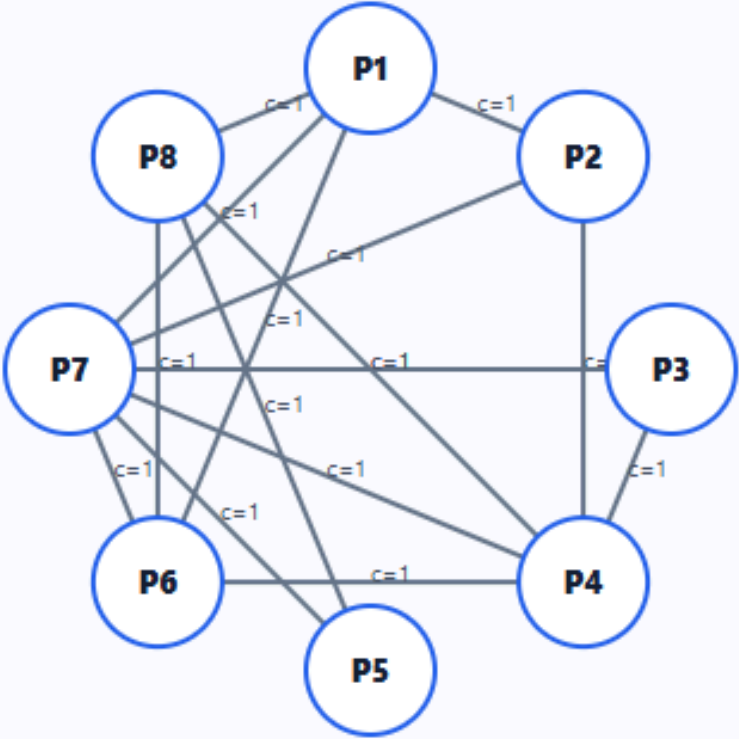


An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale → Deep Residual Learning for Image Recognition

02 • VISUALIZATION

Citation Network Map

The graph view makes the relationship structure easy to explain during the viva.



03 • DATA

Research Paper Dataset

The seeded records are real publications. The project uses a selected citation network for demonstration; it is not intended to reproduce the entire academic citation database.

ID	Research paper	Authors	Year	Manage
P1	ImageNet: A large-scale hierarchical image database	J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei	2009	<button>Delete</button>
P2	ImageNet Classification with Deep Convolutional Neural Networks	A. Krizhevsky, I. Sutskever, G. E. Hinton	2012	<button>Delete</button>

04 • TRAVERSAL

BFS & DFS

BFS explores level-by-level using a queue. DFS explores deeply before backtracking. Both traverse the adjacency list in $O(V + E)$.

Start paper

P7 — Swin Transformer: Hierarchical Vision Transformer✓

Run BFS

Run DFS

BFS order: P7 → P1 → P2 → P3 → P4 → P5 → P6

05 • PATH ANALYSIS

Citation Path

Find a directed path from one paper to another using BFS and parent tracking.

Source / citing paper

P7 — Swin Transformer: Hierarchical Vision Transformer✓

Target / referenced paper

P1 — ImageNet: A large-scale hierarchical image database✓

Find Path

Path: P7 → P1

Edges traversed: 1

06 • NETWORK FLOW

Maximum Citation Flow

Uses **Edmonds-Karp**, a Ford-Fulkerson implementation that uses BFS to find augmenting paths in the residual graph.

Source

P7 — Swin Transformer: Hierarchical Vision Transformer✓

Target

P1 — ImageNet: A large-scale hierarchical image database✓

Calculate Maximum Flow

3

Maximum modeled flow: P7 → P1

1. P7 → P1 bottleneck 1
2. P7 → P2 → P1 bottleneck 1
3. P7 → P6 → P1 bottleneck 1

Capacity is a project-defined flow unit, not a Google Scholar citation count.

07 • RANKING

Influential Papers

The project ranks papers by **in-degree**, i.e. the number of incoming citation edges inside this selected network. A Java **PriorityQueue** produces the ranking.

Rank	ID	Paper	Incoming citations
#1	P1	ImageNet: A large-scale hierarchical image database	4
#2	P4	Deep Residual Learning for Image Recognition	3
#3	P2	ImageNet Classification with Deep Convolutional Neural Networks	2
#4	P3	Very Deep Convolutional Networks for Large-Scale Image Recognition	2
#5	P5	Attention Is All You Need	2
#6	P6	An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale	2
#7	P7	Swin Transformer: Hierarchical Vision Transformer using Shifted Windows	0
#8	P8	Learning Transferable Visual Models From Natural Language Supervision	0

08 • MANAGEMENT

Add Research Paper

Adds a new graph vertex to the current Java session.

Demo: use P9, then create P9 → P7 in the citation form.

ID

P9

Title

09 • RELATIONSHIP

Create Citation

From = citing paper → **To** = referenced paper. Capacity must be a positive integer.

From

P8 — Learning Transferable Visual Models From Natural Language Supervision



To

P7 — Swin Transformer: Hierarchical Vision Transformer using Shifted Windows



10 • COMPLEXITY

Algorithm & Scalability Analysis

These complexity values directly support the computational-requirements discussion in the DSA PPT.

BFS

$O(V + E)$

Queue + visited set

DFS

$O(V + E)$

Recursion + visited set

Path BFS

$O(V + E)$

Parent reconstruction

Ranking

$O(E + V \log V)$

Indegree + PriorityQueue

Edmonds-Karp

$O(VE^2)$

Residual BFS iterations

11 • REPORT

Generated Citation Analysis Report

Current network: 8 papers, 15 directed citation edges.

Top in-degree paper: P1 — ImageNet: A large-scale hierarchical image database (4 incoming edges).

Demo flow: $P7 \rightarrow P1 = 3$ modeled flow units.

Algorithms: Adjacency List • BFS • DFS • BFS Path • PriorityQueue Ranking • Edmonds-Karp Maximum Flow.

Complexities: BFS/DFS/Path $O(V+E)$; Ranking $O(E+V \log V)$; Edmonds-Karp $O(VE^2)$.

Interpretation: this system demonstrates algorithmic citation-network analysis on a selected real-paper dataset. Edge capacity and in-degree are project metrics, not authoritative academic impact metrics.

Download Report (.txt)

FUTURE SCOPE

- Real-time research paper data – Connect the system with research databases to automatically obtain updated papers and citation relationships.
- Large-scale citation analysis – Extend the system to handle thousands or millions of research papers efficiently.
- Advanced ranking algorithms – Implement algorithms such as PageRank to identify influential research papers more effectively.
- AI-based research insights – Use AI/ML to identify research trends and recommend related papers.
- Interactive visualization and cloud deployment – Provide advanced graph visualization and deploy the system as an online web application.

THANK YOU
